**University of Florida, Department of Mathematics**

# Algebra Qualifying Exam, January 2026, Part 1

*Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.*

**1.** Let $n \geq 2$ and let $G$ be a cyclic group of order $n$. Prove that $\mathrm{Aut}(G) \cong (\mathbb{Z}/n\mathbb{Z})^\times$. (Here $(\mathbb{Z}/n\mathbb{Z})^\times$ is the unit group of the ring $\mathbb{Z}/n\mathbb{Z}$.)

**2.** Prove that there is no simple group of order $9477 = 3^6 \cdot 13$. (Consider the action of $G$ on the set of left cosets of a Sylow 3-subgroup of $G$.)

**3.** Construct four groups of order 28, and prove that they are pairwise non-isomorphic.

**4.** Let $G$ be a group.

(a) Define the upper central series of $G$:

$$Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \cdots$$

(b) Prove that the groups $Z_i(G)$ are all characteristic subgroups of $G$.

(c) Give an example of a nontrivial group $G$ such that the groups $Z_i(G)$ are all trivial.

**5.** Prove that the ring $\mathbb{Z}[\sqrt{3}] = \{a + b\sqrt{3} : a, b \in \mathbb{Z}\}$ is a Euclidean domain with respect to the norm $N(a + b\sqrt{3}) = |a^2 - 3b^2|$.

**6.** For each of the following polynomials $f(X) \in \mathbb{Q}[X]$, either prove that $f(X)$ is irreducible or give a nontrivial factorization of $f(X)$.

(a) $f(X) = X^4 - 6X^2 + 30X - 90$

(b) $f(X) = X^4 + 5X^3 - 6X^2 - 23$

(c) $f(X) = X^4 + X^3 + X^2 + X + 1$